RESULT:

========== QUANTIZATION RESULTS ==========
Amplitude = 1.0 | Frequency = 50 Hz | Fs = 1000 Hz
Samples = 1000 | Main word length = 8 bits
8-bit Step Size = 0.0078125
Rounding SQNR = 48.9339 dB
Truncation SQNR = 45.2931 dB
Output SQNR = 48.9339 dB
Max Rounding Error = 0.003483
Max Truncation Error = 0.005744

FIRST 10 SAMPLES

n Original Rounded RoundError Truncated TruncError

_ ———— ———— ———— ———— ————

0 0 0 0 0 0
1 0.30902 0.3125 -0.003483 0.30469 0.0043295
2 0.58779 0.58594 0.0018478 0.58594 0.0018478
3 0.80902 0.8125 -0.003483 0.80469 0.0043295
4 0.95106 0.95312 -0.0020685 0.94531 0.005744
5 1 1 0 1 0
6 0.95106 0.95312 -0.0020685 0.94531 0.005744
7 0.80902 0.8125 -0.003483 0.80469 0.0043295
8 0.58779 0.58594 0.0018478 0.58594 0.0018478
9 0.30902 0.3125 -0.003483 0.30469 0.0043295

WORD-LENGTH COMPARISON
Bits StepSize SQNR_Round_dB SQNR_Trunc_dB MaxError_Round MaxError_Trunc

——— ———— ———— ———— ———— ————

4 0.125 23.667 20.871 0.059017 0.087785
8 0.0078125 48.934 45.293 0.003483 0.005744
12 0.00048828 78.811 65.891 0.00011536 0.00042324
16 3.0518e-05 99.697 91.599 1.382e-05 2.6516e-05

OUTPUT PARAMETER TABLE
Parameter Observation Unit

———————————— ————————————————————————————————
————————

{'Quantized signal' }{'8-bit rounding, Delta = 0.0078125' }{'Amplitude'}
{'Quantization error' }{'Max rounding error = 0.003483' }{'Amplitude'}
{'SQNR' }{'Rounding = 48.9339 dB; Truncation = 45.2931 dB'}{'dB' }
{'Word-length comparison'}{'SQNR increases as word length increases' }{'Bits / dB'}

========== FINAL RESULT ==========
Uniform quantization using rounding and truncation was performed.
Quantization error and SQNR were calculated.
Increasing word length reduces quantization error and improves SQNR.
Finite word-length effects are therefore verified.

>> ========== QUANTIZATION RESULTS ==========

Amplitude = 1.0 | Frequency = 50 Hz | Fs = 1000 Hz

Samples = 1000 | Main word length = 8 bits

8-bit Step Size = 0.0078125

Rounding SQNR = 48.9339 dB

Truncation SQNR = 45.2931 dB

Output SQNR = 48.9339 dB

Max Rounding Error = 0.003483

Max Truncation Error = 0.005744

FIRST 10 SAMPLES

n	Original	Rounded	RoundError	Truncated	TruncError
0	0	0	0	0	0
1	0.30902	0.3125	-0.003483	0.30469	0.0043295
2	0.58779	0.58594	0.0018478	0.58594	0.0018478
3	0.80902	0.8125	-0.003483	0.80469	0.0043295
4	0.95106	0.95312	-0.0020685	0.94531	0.005744
5	1	1	0	1	0

6	0.95106	0.95312	-0.0020685	0.94531	0.005744
7	0.80902	0.8125	-0.003483	0.80469	0.0043295
8	0.58779	0.58594	0.0018478	0.58594	0.0018478
9	0.30902	0.3125	-0.003483	0.30469	0.0043295

WORD-LENGTH COMPARISON

Bits	StepSize	SQNR_Round_dB	SQNR_Trunc_dB	MaxError_Round	MaxError_Trunc
4	0.125	23.667	20.871	0.059017	0.087785
8	0.0078125	48.934	45.293	0.003483	0.005744
12	0.00048828	78.811	65.891	0.00011536	0.00042324
16	3.0518e-05	99.697	91.599	1.382e-05	2.6516e-05

OUTPUT PARAMETER TABLE

Parameter	Observation	Unit
{'Quantized signal'}	{'8-bit rounding, Delta = 0.0078125'}	{'Amplitude'}
{'Quantization error'}	{'Max rounding error = 0.003483'}	{'Amplitude'}
{'SQNR'}	{'Rounding = 48.9339 dB; Truncation = 45.2931 dB'}	{'dB'}
{'Word-length comparison'}	{'SQNR increases as word length increases'}	{'Bits / dB'}

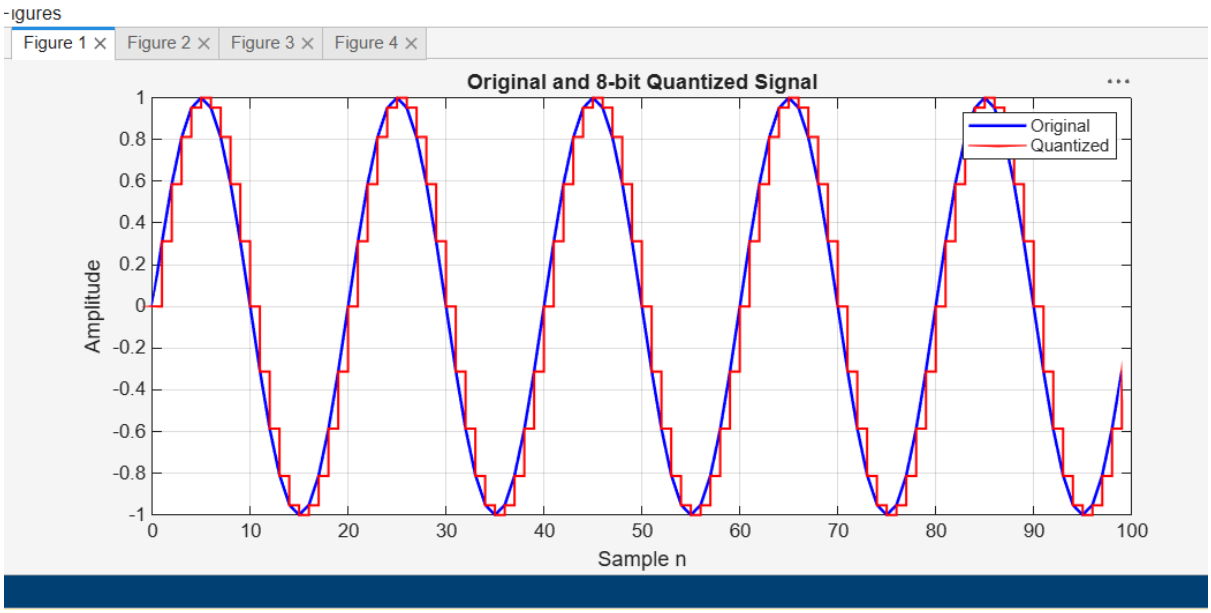
========== FINAL RESULT ==========

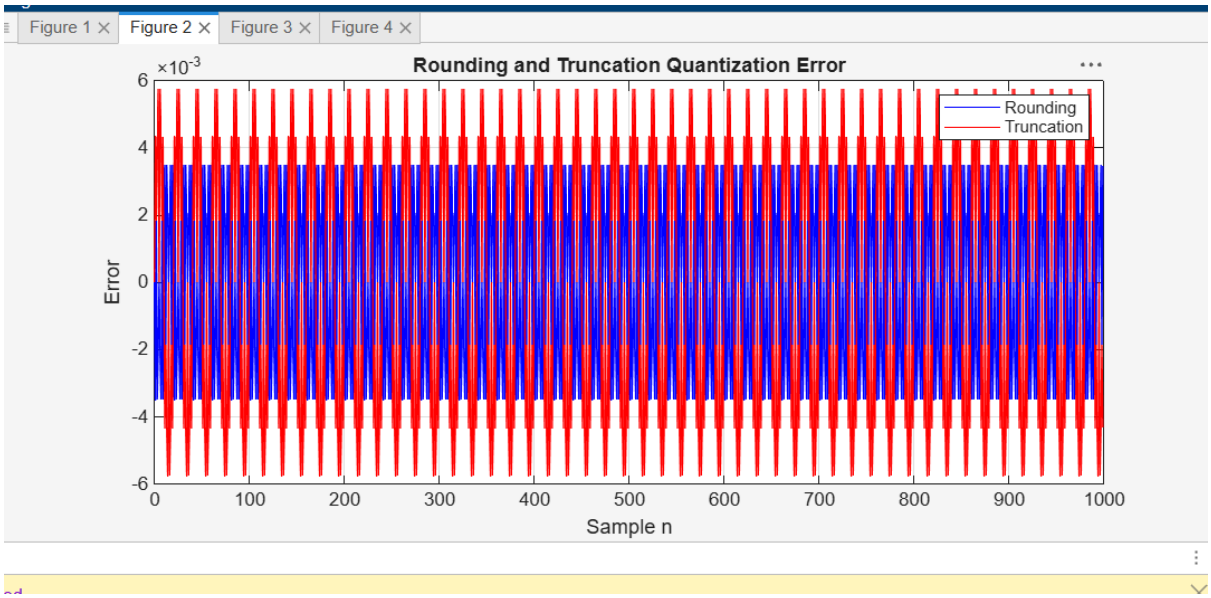
Uniform quantization using rounding and truncation was performed.

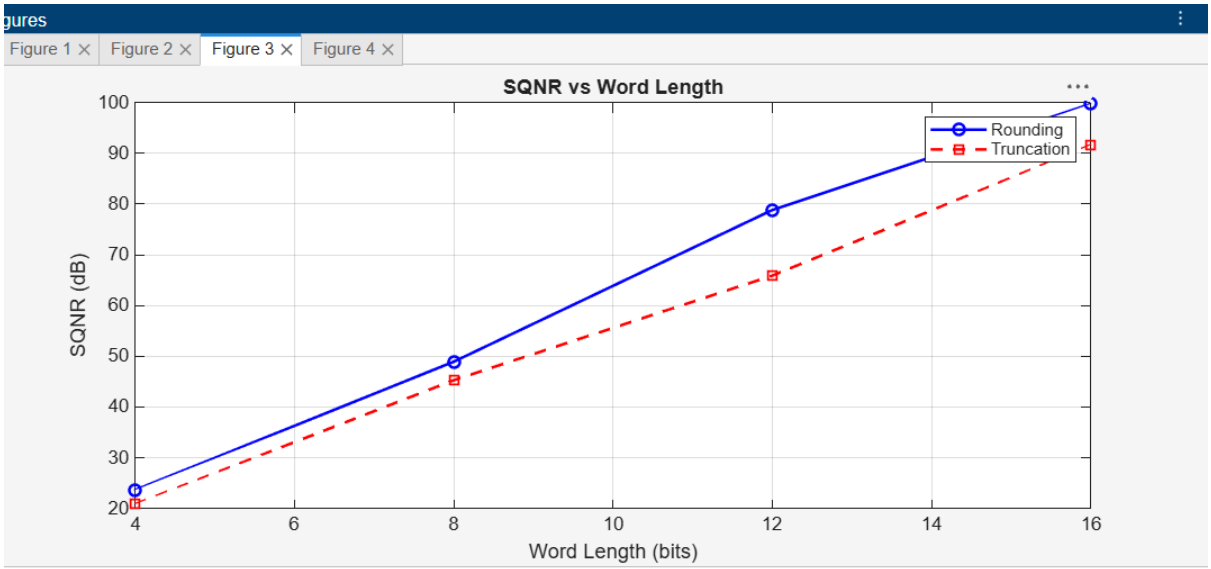
Quantization error and SQNR were calculated.

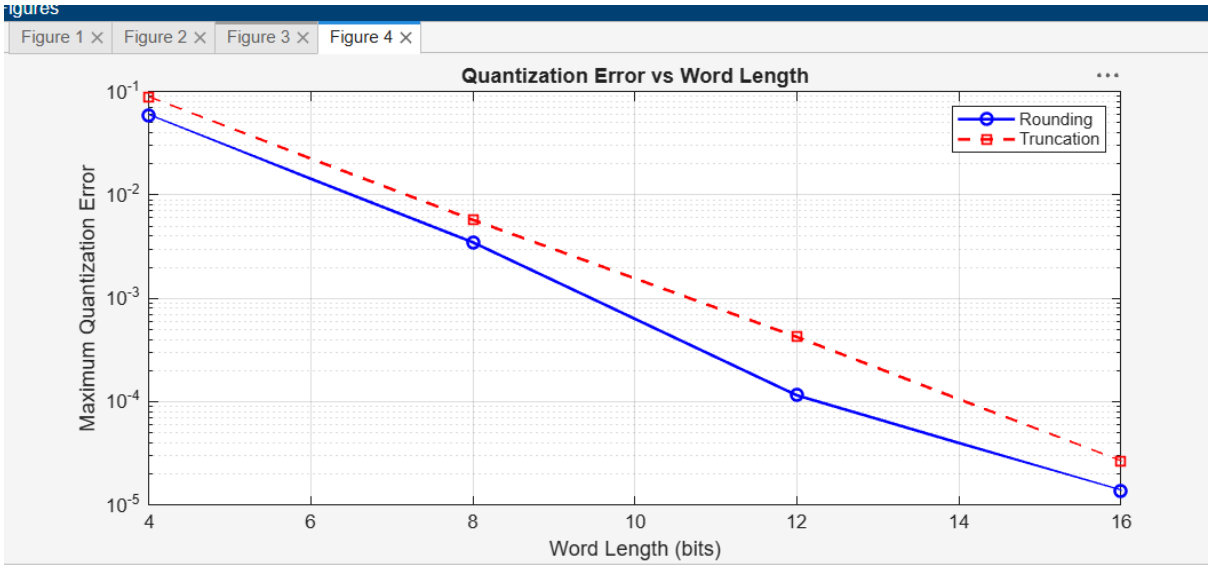
Increasing word length reduces quantization error and improves SQNR.

Finite word-length effects are therefore verified.






Figure 1
Figure 2
Figure 3
Figure 4
SQNR vs Word Length
SQNR (dB)
Word Length (bits)
Rounding
Truncation
100
90
80
70
60
50
40
30
20
4
6
8
10
12
14
16


Figure 1
Figure 2
Figure 3
Figure 4
Quantization Error vs Word Length
Maximum Quantization Error
Word Length (bits)
Rounding
Truncation
10^{-1}
10^{-2}
10^{-3}
10^{-4}
10^{-5}
4
6
8
10
12
14
16